

ROCHES POINT, Ireland

WMO: 039520

Lat: 51.800N

Lon: 8.250W

Elev: 135

StdP: 14.62

Time Zone: 1.00 (IRL)

Period: 01-15

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	31.8	34.1	25.4	19.5	34.6	27.7	21.7	36.6	44.8	47.8	40.5	47.3	10.4	350	0.740

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	9.0	68.2	62.1	66.3	60.8	64.8	59.8	63.7	66.5	62.4	64.6	61.3	63.5	11.9	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.5	84.9	64.7	61.4	81.7	63.5	60.3	78.5	62.4	29.0	66.7	28.1	64.7	27.4	63.6	69.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 36.4	32.0	28.2	30.3	73.2	2.9	2.4	28.2	74.9	26.5	76.3	24.9	77.7	22.8	79.4		
(5)			28.8	66.7	2.8	1.9	26.8	68.1	25.1	69.2	23.5	70.3	21.5	71.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	51.3	44.5	44.2	45.5	48.7	52.2	57.0	59.5	59.6	57.5	53.5	48.4	45.1
	DBStd	6.80	4.33	4.02	3.74	3.27	2.96	3.08	3.19	2.45	3.02	3.82	4.30	4.70
	HDD50	804	172	165	143	61	12	0	0	0	0	15	77	159
	HDD65	4987	634	583	603	489	398	241	172	168	225	358	498	618
	CDD50	1294	3	2	4	21	79	209	296	298	225	122	29	6
	CDD65	3	0	0	0	0	0	0	2	1	0	0	0	0
	CDH74	1	0	0	0	0	0	0	1	0	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	14.0	16.0	15.0	14.4	13.5	14.1	12.3	12.0	12.8	12.4	14.2	15.1	15.8
Precipitation	PrecAvg	37.2	4.1	3.4	3.0	2.2	2.5	2.1	2.2	3.2	3.1	3.8	3.6	3.9
	PrecMax	48.5	8.7	8.1	6.3	7.0	5.6	5.6	4.8	9.4	6.9	8.1	6.9	7.4
	PrecMin	27.0	1.0	0.2	0.1	0.6	0.5	0.6	0.2	0.7	0.2	0.9	1.5	0.6
	PrecStd	5.0	1.9	2.0	1.7	1.4	1.4	1.2	1.3	1.8	1.8	2.0	1.7	1.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.4	54.3	56.4	60.7	64.5	68.4	72.3	70.2	67.0	62.7	58.2	54.8
		MCWB	52.3	51.9	50.7	52.5	57.8	60.8	64.7	63.8	61.6	58.5	56.3	53.3
	2%	DB	52.9	52.5	53.8	57.0	61.1	65.9	69.0	67.3	65.1	60.9	56.6	53.3
		MCWB	51.2	50.6	49.6	51.5	55.4	59.6	62.8	61.4	60.4	58.3	54.9	51.8
	5%	DB	51.8	51.2	52.0	55.1	58.9	63.9	66.8	65.6	63.6	59.8	55.5	52.5
		MCWB	50.3	49.3	48.6	51.0	53.8	58.5	61.5	60.2	59.7	57.6	54.0	51.1
	10%	DB	50.8	49.9	50.7	53.6	57.1	62.3	64.9	64.1	62.3	58.8	54.5	51.5
		MCWB	49.4	48.2	48.2	50.3	52.8	57.5	60.3	59.4	59.2	56.9	52.8	50.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	53.1	52.8	52.3	54.9	59.6	62.3	65.7	65.5	63.4	60.4	57.1	54.0
		MCDB	53.7	53.7	54.5	57.3	62.8	66.0	69.8	68.6	65.0	60.9	58.0	54.3
	2%	WB	51.9	51.1	50.9	53.3	56.7	60.9	63.9	63.5	62.0	59.4	55.5	52.6
		MCDB	52.4	52.0	52.2	55.2	59.5	64.4	67.3	65.6	63.4	60.1	56.1	53.1
	5%	WB	50.9	49.8	49.8	52.1	55.1	59.7	62.7	62.0	61.1	58.3	54.5	51.4
		MCDB	51.5	50.8	50.9	53.9	57.4	62.7	65.6	63.8	62.6	59.1	55.3	52.1
	10%	WB	49.8	48.6	48.9	51.1	54.0	58.6	61.2	61.1	60.1	57.4	53.2	50.5
		MCDB	50.5	49.7	50.3	52.9	56.0	61.3	64.0	62.8	61.7	58.4	54.3	51.4
Mean Daily Temperature Range	5% DB	MDBR	7.1	7.2	8.3	8.9	9.0	9.4	9.1	9.0	8.6	7.6	7.3	6.5
		MCDBR	7.1	7.2	9.0	10.1	11.4	11.3	11.2	10.7	9.4	7.5	7.2	6.6
		MCWBR	7.6	6.6	7.1	7.0	7.3	7.2	6.8	6.8	6.8	6.4	7.0	7.2
	5% WB	MCDBR	7.0	6.5	7.7	8.7	9.1	9.9	10.0	8.7	8.2	6.5	6.7	6.5
		MCWBR	7.8	6.5	6.9	7.0	6.8	6.9	6.8	6.9	7.0	6.4	7.2	7.2
Clear-Sky Solar Irradiance	taub	0.338	0.348	0.372	0.377	0.388	0.372	0.378	0.360	0.357	0.349	0.331	0.329	
	taud	2.293	2.352	2.307	2.333	2.335	2.411	2.399	2.462	2.465	2.441	2.362	2.263	
	Ebn at Noon	205	243	259	273	274	280	276	276	263	240	208	185	
	Edh at Noon	21	26	33	36	37	35	35	31	28	24	20	19	
All-Sky Solar Radiation	RadAvg	247	468	831	1333	1624	1768	1605	1364	992	569	309	188	
	RadStd	24	36	83	98	99	143	131	80	74	52	23	23	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (10 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page